Stochastic Calculus

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1 Random walk

1.1 Symmetric Random Walk

We model repeated fair coin tosses by an infinite sequence $\omega = \omega_1\omega_2\dots$, where $\omega_j \in \{H, T\}$ and $\mathbb{P}(H) = \mathbb{P}(T) = \frac{1}{2}$. Define $M_0 = 0$ and

$$X_j(\omega) = \begin{cases} +1, & \omega_j = H, \\ -1, & \omega_j = T, \end{cases} \qquad j \geq 1$$

$$M_k = \sum_{j=1}^{k} X_j, \quad k \geq 1.$$

The process $(M_k)_{k\geq 0}$ is called the symmetric random walk. At each step it moves up or down by one unit, each with probability $\frac{1}{2}$.

For integers $0 \leq k < l$,

$$M_l - M_k = \sum_{j=k+1}^{l} X_j.$$

Because the X_j are independent, increments over disjoint time intervals are independent. Moreover, $\mathbb{E}(M_l - M_k) = 0$ and $\mathrm{Var}(M_l - M_k) = \sum_{j=k+1}^{l} \mathrm{Var}(X_j) = l - k$.

1.2 Martingale Property

Let $\mathcal{F}_k = \sigma(X_1, \dots, X_k)$ be the information after k tosses. Then for $l > k$

$$\mathbb{E}[M_l|\mathcal{F}_k] = \mathbb{E}[(M_l - M_k) + M_k|\mathcal{F}_k] = \mathbb{E}[M_l - M_k] + M_k = M_k,$$

since $M_l - M_k$ depends only on future tosses and has mean 0. Hence $(M_k, \mathcal{F}_k)$ is a martingale.

1.3 Quadratic Variation

Define the quadratic variation up to time k by

$$[M, M]_k = \sum_{j=1}^{k} (M_j - M_{j-1})^2.$$

Here $M_j - M_{j-1} = X_j \in \{\pm 1\}$, so $(M_j - M_{j-1})^2 = 1$ and therefore

$$[M, M]_k = \sum_{j=1}^{k} 1 = k.$$

This is a pathwise computation: along every realized path, the quadratic variation equals the elapsed time.

1.4 The natural filtration of the random walk

Recall that the underlying sample space is

$$\Omega = \{H, T\}^{\mathbb{N}} = \{\omega = (\omega_1, \omega_2, \dots) : \omega_j \in \{H, T\}\}.$$

and that $X_j(\omega) = +1$ if $\omega_j = H$ and -1 if $\omega_j = T$. For each $k \geq 0$, define

$$\mathcal{F}_k = \sigma(X_1, \dots, X_k) = \sigma(M_0, M_1, \dots, M_k)$$

This is called the natural filtration of the random walk. The σ-algebra $\mathcal{F}_k$ represents all information revealed by the first k coin tosses. Every event in $\mathcal{F}_k$ is a union of sets of the form $\{\omega : X_1(\omega) = \epsilon_1, \dots, X_k(\omega) = \epsilon_k\}$ where $\epsilon_j \in \{-1, +1\}$.

Example 1.1 (The first two σ-algebras)**.** *For $k = 1$,*

$$\mathcal{F}_1 = \{\emptyset, \{X_1 = +1\}, \{X_1 = -1\}, \Omega\} = \{\emptyset, \{M_1 = 1\}, \{M_1 = -1\}, \Omega\}.$$

For $k = 2$, the four basic sets are

$$\begin{aligned}
A_{++} &= \{X_1 = +1, X_2 = +1\} = \{M_1 = 1, M_2 = 2\} \\
A_{+-} &= \{X_1 = +1, X_2 = -1\} = \{M_1 = 1, M_2 = 0\} \\
A_{-+} &= \{X_1 = -1, X_2 = +1\} = \{M_1 = -1, M_2 = 0\} \\
A_{--} &= \{X_1 = -1, X_2 = -1\} = \{M_1 = -1, M_2 = -2\}
\end{aligned}$$

Thus $\mathcal{F}_2 = \{$all unions of $A_{++}, A_{+-}, A_{-+}, A_{--}\}$, which has $2^4 = 16$ elements. Moreover, $\{X_1 = +1\} = A_{++} \cup A_{+-} \in \mathcal{F}_2$ and $\{X_1 = -1\} = A_{-+} \cup A_{--} \in \mathcal{F}_2$, so $\mathcal{F}_1 \subset \mathcal{F}_2$. In general,

$$\mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2 \subset \dots,$$

and $(\mathcal{F}_k)_{k \geq 0}$ is a discrete-time filtration.

Remark 1.2. *An event lies in $\mathcal{F}_k$ if and only if it can be decided by observing the random walk up to time k. Events depending on future steps, such as $\{M_{k+1} = M_k + 1\}$ or "the walk ever hits +10", are not in $\mathcal{F}_k$.*

1.5 Scaled Random Walk

To approximate Brownian motion, we speed up time and scale down space. Fix $n \in \mathbb{N}$ and define

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} M_{\lfloor nt \rfloor}, \quad t \geq 0,$$

with linear interpolation between grid points. Then for $0 \leq s < t$ with $ns, nt \in \mathbb{N}$,

$$\mathbb{E}(W^{(n)}(t) - W^{(n)}(s)) = 0, \quad \mathrm{Var}(W^{(n)}(t) - W^{(n)}(s)) = t - s.$$

Moreover, increments over disjoint intervals are independent. Thus the scaled walk has the same mean-variance structure as Brownian increments. Its quadratic variation satisfies, pathwise,

$$[W^{(n)}, W^{(n)}](t) = \sum_{j=1}^{\lfloor nt \rfloor} \left(W^{(n)}\left(\frac{j}{n}\right) - W^{(n)}\left(\frac{j-1}{n}\right)\right)^2 = \sum_{j=1}^{\lfloor nt \rfloor} \frac{1}{n} \approx t.$$

1.6 Central Limit and the Brownian Limit

For fixed $t \geq 0$,

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} \sum_{j=1}^{\lfloor nt \rfloor} X_j.$$

By the Central Limit Theorem,

$$W^{(n)}(t) \to \mathcal{N}(0, t).$$

More generally, the finite-dimensional distributions of $\{W^{(n)}(t)\}_{t \geq 0}$ converge to those of a process with independent, stationary Normal increments of variance $t - s$ over $[s, t]$.

2 Quadratic Variation

Fix a time horizon $T > 0$ and consider a partition,

$$\Pi = \{0 = t_0 < t_1 < \cdots < t_n = T\}$$

with mesh size $\|\Pi\| = \max_{0 \leq j \leq n-1}(t_{j+1} - t_j)$.

2.1 Quadratic variation: deterministic functions

Definition 2.1 (Quadratic variation). *Let $f : [0, T] \to \mathbb{R}$. The quadratic variation of f up to time T is defined by*

$$[f, f](T) = \lim_{\|\Pi\| \to 0} \sum_{j=0}^{n-1} (f(t_{j+1}) - f(t_j))^2,$$

provided the limit exists.

Remark 2.2. *Suppose f has a continuous derivative on $[0,T]$. By the mean value theorem, $f(t_{j+1})-f(t_j) = f'(t_j^*)(t_{j+1}-t_j)$ for some $t_j^* \in (t_j, t_{j+1})$. Hence*

$$\sum_{j=0}^{n-1}(f(t_{j+1}) - f(t_j))^2 = \sum_{j=0}^{n-1} |f'(t_j^*)|^2(t_{j+1}-t_j)^2 \leq \|\Pi\| \sum_{j=0}^{n-1} |f'(t_j^*)|^2(t_{j+1}-t_j).$$

As $\|\Pi\| \to 0$ the second sum converges to $\int_0^T |f'(t)|^2 dt$ (a Riemann sum), which is finite by continuity of f'. Therefore

$$[f,f](T) \leq \lim_{\|\Pi\|\to 0} \|\Pi\| \int_0^T |f'(t)|^2 dt = 0.$$

So continuously differentiable functions have zero quadratic variation.

3 Brownian Motion

Definition 3.1. *W_t, a Brownian motion, is a continuous function in $[0,T]$ where:*

1. *$W_0 = 0$*
2. *W_t has independent increments, i.e., $W_t - W_s$ is independent of $W_r - W_q$, where $0 \leq q \leq r \leq s \leq t$*
3. *$W_t - W_s \sim \mathcal{N}(0, t-s)$, $t > s$.*

3.1 Filtration for Brownian motion

Definition 3.2 (Filtration for Brownian motion)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space on which a Brownian motion $W = (W(t))_{t\geq 0}$ is defined. A filtration $(\mathcal{F}(t))_{t\geq 0}$ for W is a collection of σ-algebras satisfying:*

1. *(Information accumulates) If $0 \leq s < t$, then $\mathcal{F}(s) \subseteq \mathcal{F}(t)$.*
2. *(Adaptedness) For each $t \geq 0$, $W(t)$ is $\mathcal{F}(t)$-measurable.*
3. *(Independence of future increments) For $0 \leq t < u$ the increment $W(u) - W(t)$ is independent of $\mathcal{F}(t)$.*

Definition 3.3 (Adapted process)**.** *A stochastic process $\Delta = (\Delta(t))_{t\geq 0}$ is said to be adapted to $(\mathcal{F}(t))_{t\geq 0}$ if for each $t \geq 0$, the random variable $\Delta(t)$ is $\mathcal{F}(t)$-measurable.*

3.2 Martingale property

Theorem 3.4 (Brownian motion is a martingale)**.** *Let W be a Brownian motion with respect to $(\mathcal{F}(t))_{t\geq 0}$. Then for $0 \leq s \leq t$,*

$$\mathbb{E}[W(t)|\mathcal{F}(s)] = W(s).$$

3.3 Quadratic variation of Brownian motion

Definition 3.5. *Given a partition* $\Pi = \{0 = t_0 < \cdots < t_n = T\}$*, define the sampled quadratic variation:*

$$Q_\Pi(W;T) = \sum_{j=0}^{n-1} (W(t_{j+1}) - W(t_j))^2.$$

Theorem 3.6 (Quadratic variation of Brownian motion)**.** *Let* W_t *be a Brownian motion. Then for every* $T \geq 0$

$$[W,W](T) = \lim_{\|\Pi\|\to 0} \sum_{j=0}^{n-1} (W(t_{j+1}) - W(t_j))^2 = T \quad \textit{almost surely.}$$

Proof. The increments $\Delta_j W = W(t_{j+1}) - W(t_j)$ are independent and $\Delta_j W \sim \mathcal{N}(0, t_{j+1} - t_j)$. Hence $\mathbb{E}[(\Delta_j W)^2] = t_{j+1} - t_j$ and $\mathbb{E}[(\Delta_j W)^4] = 3(t_{j+1} - t_j)^2$ (the fourth moment of a centered normal equals 3 times the variance squared). Therefore

$$\mathbb{E}[Q_\Pi] = \sum_{j=0}^{n-1} \mathbb{E}[(\Delta_j W)^2] = \sum_{j=0}^{n-1} (t_{j+1} - t_j) = T.$$

Next compute the variance. Since the summands are independent,

$$\mathrm{Var}(Q_\Pi) = \sum_{j=0}^{n-1} \mathrm{Var}((\Delta_j W)^2).$$

Using $\mathrm{Var}((\Delta_j W)^2) = \mathbb{E}[(\Delta_j W)^4] - (\mathbb{E}[(\Delta_j W)^2])^2 = 3(t_{j+1} - t_j)^2 - (t_{j+1} - t_j)^2 = 2(t_{j+1} - t_j)^2$, we obtain

$$\mathrm{Var}(Q_\Pi) = 2\sum_{j=0}^{n-1} (t_{j+1} - t_j)^2 \leq 2\|\Pi\| \sum_{j=0}^{n-1} (t_{j+1} - t_j) = 2\|\Pi\| T.$$

Hence $\mathrm{Var}(Q_\Pi) \to 0$ as $\|\Pi\| \to 0$. Since $\mathbb{E}[(Q_\Pi - T)^2] = \mathrm{Var}(Q_\Pi)$, we conclude that $Q_\Pi \to T$ in L^2. This proves the quadratic-variation limit. ☐

Remark 3.7. *Unlike* C^1 *functions (which have zero quadratic variation), Brownian motion accumulates quadratic variation at rate one per unit time. This is a fundamental reason why stochastic calculus differs from ordinary calculus, and it is the mathematical source of volatility in diffusion models for asset prices.*

Remark 3.8. *One as a mnemonic writes* $dW(t)dW(t) = dt$*, meaning that when one computes quadratic variation, the squared increment behaves like time,* $\sum(\Delta W)^2 \approx \sum \Delta t = T$*. This identity should not be interpreted as an ordinary product of differentials, it is a shorthand for the quadratic variation limit above.*

3.4 Volatility of Geometric Brownian Motion

Let $\alpha \in \mathbb{R}$ and $\sigma > 0$ be constants. Define the geometric Brownian motion

$$S(t) = S(0) \exp\left\{\sigma W(t) + \left(\alpha - \frac{1}{2}\sigma^2\right) t\right\}, \quad t \geq 0.$$

This is the asset-price model used in the Black-Scholes-Merton framework. Fix $0 \leq T_1 < T_2$ and suppose we observe $S(t)$ for $t \in [T_1, T_2]$. Choose a partition $T_1 = t_0 < t_1 < \cdots < t_m = T_2$, and consider the log-returns,

$$\log \frac{S(t_{j+1})}{S(t_j)} = \sigma(W(t_{j+1}) - W(t_j)) + \left(\alpha - \frac{1}{2}\sigma^2\right)(t_{j+1} - t_j), \quad j = 0, \ldots, m-1.$$

The sum of squared log-returns (often called the realized volatility) is

$$\sum_{j=0}^{m-1}\left(\log \frac{S(t_{j+1})}{S(t_j)}\right)^2 = \sigma^2 \sum_{j=0}^{m-1}(W(t_{j+1}) - W(t_j))^2 + \left(\alpha - \frac{1}{2}\sigma^2\right)^2 \sum_{j=0}^{m-1}(t_{j+1} - t_j)^2 + 2\sigma\left(\alpha - \frac{1}{2}\sigma^2\right)\sum_{j=0}^{m-1}(W(t_{j+$$

As $\|\Pi\| \to 0$,

$$\sum_{j=0}^{m-1}(W(t_{j+1}) - W(t_j))^2 \longrightarrow T_2 - T_1, \quad \sum_{j=0}^{m-1}(t_{j+1} - t_j)^2 \longrightarrow 0, \quad \sum_{j=0}^{m-1}(W(t_{j+1}) - W(t_j))(t_{j+1} - t_j) \longrightarrow 0.$$

Therefore, for fine partitions ($\|\Pi\| \to 0$)

$$\sum_{j=0}^{m-1}\left(\log \frac{S(t_{j+1})}{S(t_j)}\right)^2 \approx \sigma^2(T_2 - T_1), \quad \text{or equivalently,} \quad \frac{1}{T_2 - T_1}\sum_{j=0}^{m-1}\left(\log \frac{S(t_{j+1})}{S(t_j)}\right)^2 \approx \sigma^2.$$

3.5 Conditional expectation with Brownian motion

Let $(W_t)_{t \geq 0}$ be a standard Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$ with its natural filtration $\mathcal{F}_t = \sigma(W_u : 0 \leq u \leq t)$. Fix $0 \leq s \leq t$.

Trading gains and the Itô integral Fix a time horizon $T > 0$ and a partition (trading dates) $\Pi = \{0 = t_0 < t_1 < \cdots < t_n = T\}$. A trader chooses, at each trading time t_j, a position $\Delta(t_j)$ (number of shares) and then holds it constant until the next trading time and then sells it off. Formally, we model the position process by a simple adapted process:

$$\Delta(t) = \sum_{j=0}^{n-1} \Delta(t_j) \mathbf{1}_{[t_j, t_{j+1})}(t),$$

where each $\Delta(t_j)$ is $\mathcal{F}_{t_j}$-measurable (i.e. depends only on information available up to time t_j). This is the non-anticipativity condition that you cannot choose your position using future price increments.

Consider what happens on a single interval $[t_j, t_{j+1})$. If you hold $\Delta(t_j)$ shares throughout this interval, then your gain over the interval is

$$\text{gain on } [t_j, t_{j+1}) = \Delta(t_j)(W(t_{j+1}) - W(t_j)).$$

Summing over intervals, the total gain from 0 to T is

$$G_T(\Delta; \Pi) = \sum_{j=0}^{n-1} \Delta(t_j)(W(t_{j+1}) - W(t_j)).$$

More generally, for an intermediate time $t \in [t_k, t_{k+1})$ the gain up to time t is

$$G_t(\Delta; \Pi) = \sum_{j=0}^{k-1} \Delta(t_j)(W(t_{j+1}) - W(t_j)) + \Delta(t_k)(W(t) - W(t_k)).$$

The choice $\Delta(t_j)$ at time t_j must be based on $\mathcal{F}_{t_j}$, hence it must be independent of the future increment $W(t_{j+1}) - W(t_j)$. This matches the economic interpretation: you decide how many shares to hold before the price move on $(t_j, t_{j+1}]$ is revealed. This is exactly why the sum uses the left endpoint t_j. If we instead used $\Delta(t_{j+1})$ (a right endpoint), then the strategy could depend on $W(t_{j+1})$, i.e. it could "peek into the future", which is not an admissible trading rule.

Definition 3.9 (Itô integral for a simple process)**.** *Let Δ be a simple adapted process of the form $\Delta(t) = \sum_{j=0}^{n-1} \Delta(t_j)\mathbf{1}_{[t_j, t_{j+1})}(t)$. We define its Itô integral with respect to Brownian motion by*

$$\int_0^T \Delta(t)dW(t) = \sum_{j=0}^{n-1} \Delta(t_j)(W(t_{j+1}) - W(t_j)),$$

and for $t \in [t_k, t_{k+1})$,

$$\int_0^t \Delta(u)dW(u) = \sum_{j=0}^{k-1} \Delta(t_j)(W(t_{j+1}) - W(t_j)) + \Delta(t_k)(W(t) - W(t_k)).$$

Remark 3.10 (Interpretation)**.** *The process $(\int_0^t \Delta(u)dW(u))_{0 \le t \le T}$ is exactly the cumulative trading gain when you hold $\Delta(t)$ shares of the asset with price $W(t)$.*

3.6 The Itô integral is a martingale

Assume W is a Brownian motion with respect to $(\mathcal{F}_t)_{t \ge 0}$ and Δ is simple and adapted as above. Then $(I(t))_{0 \le t \le T}$ is a martingale, i.e. for $0 \le s \le t \le T$, $\mathbb{E}[I(t)|\mathcal{F}_s] = I(s)$.

Proof. Fix $0 \le s \le t \le T$. Choose l, k such that $s \in [t_l, t_{l+1})$ and $t \in [t_k, t_{k+1})$ with $l \le k$. Write

$$I(t) = \underbrace{\sum_{j=0}^{l-1} \Delta(t_j)(W(t_{j+1}) - W(t_j))}_{(A)} + \underbrace{\Delta(t_l)(W(s) - W(t_l))}_{(B)}$$

$$+ \underbrace{\Delta(t_l)(W(t_{l+1}) - W(s))}_{(C)} + \underbrace{\sum_{j=l+1}^{k-1} \Delta(t_j)(W(t_{j+1}) - W(t_j))}_{(D)} + \underbrace{\Delta(t_k)(W(t) - W(t_k))}_{(E)}.$$

Now condition on $\mathcal{F}_s$. Every term in (A) and (B) is $\mathcal{F}_s$-measurable, hence $\mathbb{E}[(A)+(B)|\mathcal{F}_s] = (A) + (B) = I(s)$. One can finish the proof by showing $\mathbb{E}[(C)+(D)+(E)|\mathcal{F}_s] = 0$. □

4 Itô Integral for General Integrands

We now extend the Itô integral from simple adapted integrands to more general stochastic processes.

Definition 4.1. *A stochastic process $\Delta(t)$, $0 \le t \le T$ is said to be an admissible integrand if:*

1. *$\Delta(t)$ is adapted to $\mathcal{F}(t)$*
2. *$\mathbb{E}\left[\int_0^T \Delta(t)^2 dt\right] < \infty$.*

The second condition ensures that the total "risk exposure" of the strategy is finite.

Let $\Delta(t)$ be an admissible integrand. We approximate Δ by a sequence of simple adapted processes $\{\Delta_n(t)\}_{n\ge1}$ of the form

$$\Delta_n(t) = \sum_{j=0}^{k_n-1} \Delta(t_j^{(n)}) \mathbf{1}_{[t_j^{(n)}, t_{j+1}^{(n)})}(t),$$

where $\Pi_n = \{0 = t_0^{(n)} < \cdots < t_{k_n}^{(n)} = T\}$ is a partition of $[0, T]$ and $\|\Pi_n\| \to 0$. We require convergence in mean square:

$$\lim_{n\to\infty} \mathbb{E}\left[\int_0^T (\Delta_n(t) - \Delta(t))^2 dt\right] = 0.$$

Definition 4.2. *Let $\Delta(t)$ be an admissible integrand and $\Delta_n(t)$ a sequence of simple processes converging to Δ in $L^2([0, T] \times \Omega)$. We define*

$$\int_0^T \Delta(t) dW(t) = \lim_{n\to\infty} \int_0^T \Delta_n(t) dW(t),$$

where the limit is taken in $L^2(\Omega)$.

4.1 Basic properties

Let $I(t) = \int_0^t \Delta(u)dW(u)$. Then the Itô integral satisfies:

1. Adaptedness: $I(t)$ is $\mathcal{F}(t)$-measurable.
2. Continuity: $t \mapsto I(t)$ has continuous sample paths.
3. Martingale property: $\mathbb{E}[I(t)|\mathcal{F}_s] = I(s)$, $s \leq t$.
4. Quadratic Variation: $[I, I](t) = \int_0^t \Delta^2(u)du$.
5. Itô's Isometry: $\mathbb{E}[(I(t))^2] = \mathbb{E}\left[\int_0^t \Delta^2(u)du\right]$.

Example 4.3 (Computing $\int_0^T W(t)dW(t)$ via simple-process approximation)**.** *Fix $T > 0$. Let $\Pi_n = \{t_j = jT/n : j = 0, 1, \dots, n\}$ be the uniform partition and define $W_j = W(t_j) = W(\frac{jT}{n})$. Approximate the integrand $\Delta(t) = W(t)$ by the simple adapted process $\Delta_n(t) = W_j$ for $t_j \leq t < t_{j+1}$, $j = 0, 1, \dots, n-1$. Then $\Delta_n(t) \to W(t)$ in $L^2([0,T] \times \Omega)$ and by definition of the Itô integral,*

$$\int_0^T W(t)dW(t) = \lim_{n\to\infty} \int_0^T \Delta_n(t)dW(t) = \lim_{n\to\infty} \sum_{j=0}^{n-1} W_j(W_{j+1} - W_j).$$

To simplify the sum, use the algebraic identity

$$W_{j+1}^2 - W_j^2 = (W_{j+1} - W_j)(W_{j+1} + W_j) = 2W_j(W_{j+1} - W_j) + (W_{j+1} - W_j)^2,$$

which rearranges to

$$W_j(W_{j+1} - W_j) = \frac{1}{2}(W_{j+1}^2 - W_j^2) - \frac{1}{2}(W_{j+1} - W_j)^2.$$

Summing from $j = 0$ to $n - 1$ gives

$$\sum_{j=0}^{n-1} W_j(W_{j+1} - W_j) = \frac{1}{2}\sum_{j=0}^{n-1}(W_{j+1}^2 - W_j^2) - \frac{1}{2}\sum_{j=0}^{n-1}(W_{j+1} - W_j)^2.$$

Since $W(0) = 0$, $\sum_{j=0}^{n-1}(W_{j+1}^2 - W_j^2) = W_n^2 - W_0^2 = W(T)^2$. Hence

$$\sum_{j=0}^{n-1} W_j(W_{j+1} - W_j) = \frac{1}{2}W(T)^2 - \frac{1}{2}\sum_{j=0}^{n-1}(W_{j+1} - W_j)^2.$$

As $n \to \infty$, the quadratic-variation limit yields $\sum_{j=0}^{n-1}(W_{j+1} - W_j)^2 \longrightarrow T$. Taking limits, we conclude

$$\int_0^T W(t)dW(t) = \frac{1}{2}(W(T)^2 - T).$$

Remark 4.4. *If g were differentiable, then $\int_0^T g(t)dg(t) = \frac{1}{2}g(T)^2 - \frac{1}{2}g(0)^2$. For Brownian motion, the extra term $-\frac{1}{2}T$ appears because $[W, W](T) = T$ (nonzero quadratic variation).*

5 Itô's lemma

In ordinary calculus, if $x(t)$ is differentiable and $f \in C^2(\mathbb{R})$, then $\frac{d}{dt} f(x(t)) = f'(x(t))x'(t)$, or $df(x(t)) = f'(x(t))dx(t)$. If we tried to apply this to $x(t) = W(t)$, we would guess $df(W(t)) = f'(W(t))dW(t)$. But Brownian motion is not differentiable and has nonzero quadratic variation: $[W, W](t) = t$. This produces an extra drift term in the correct chain rule.

Theorem 5.1 (Itô's lemma: differential form)**.** *Let W be a Brownian motion and let $f \in C^2(\mathbb{R})$. Then*

$$df(W(t)) = f'(W(t))dW(t) + \frac{1}{2} f''(W(t))dt.$$

Theorem 5.2 (Itô's lemma: integral form)**.** *Let W be a Brownian motion and let $f \in C^2(\mathbb{R})$. Then for $t \geq 0$,*

$$f(W(t)) - f(W(0)) = \int_0^t f'(W(u))dW(u) + \frac{1}{2} \int_0^t f''(W(u))du.$$

Theorem 5.3 (Itô's formula for Brownian motion)**.** *Let $(W(t))_{t \geq 0}$ be a Brownian motion with filtration $(\mathcal{F}_t)$. Let $f : [0, T] \times \mathbb{R} \to \mathbb{R}$ be such that f_t, f_x, f_{xx} exist and are continuous (i.e. $f \in C^{1,2}$). Then for every $T \geq 0$,*

$$f(T, W(T)) = f(0, W(0)) + \int_0^T f_t(t, W(t))dt + \int_0^T f_x(t, W(t))dW(t) + \frac{1}{2} \int_0^T f_{xx}(t, W(t))dt.$$

Proof sketch. Fix $T > 0$ and let $\Pi = \{0 = t_0 < t_1 < \cdots < t_n = T\}$ be a partition of $[0, T]$. Write the telescoping sum

$$f(T, W(T)) - f(0, W(0)) = \sum_{j=0}^{n-1} (f(t_{j+1}, W(t_{j+1})) - f(t_j, W(t_j))).$$

For each j, apply the second-order Taylor expansion of f at $(t_j, W(t_j))$ in the two variables (t, x):

$$f(t_{j+1}, W(t_{j+1})) - f(t_j, W(t_j)) = f_t(t_j, W(t_j))\Delta t_j + f_x(t_j, W(t_j))\Delta W_j + \frac{1}{2} f_{xx}(t_j, W(t_j))(\Delta W_j)^2 + R_j, \quad (1)$$

where $\Delta t_j := t_{j+1} - t_j, \Delta W_j := W(t_{j+1}) - W(t_j)$, and R_j collects all the remaining terms:

$$R_j = f_{tx}(t_j, W(t_j))\Delta t_j \Delta W_j + \frac{1}{2} f_{tt}(t_j, W(t_j))(\Delta t_j)^2 + (\text{higher order terms}).$$

Summing (1) over $j = 0, \ldots, n-1$ gives

$$f(T, W(T)) - f(0, W(0)) = \sum_{j=0}^{n-1} f_t(t_j, W(t_j))\Delta t_j + \sum_{j=0}^{n-1} f_x(t_j, W(t_j))\Delta W_j + \frac{1}{2} \sum_{j=0}^{n-1} f_{xx}(t_j, W(t_j))(\Delta W_j)^2 + \sum_{j=0}^{n-1}$$

By continuity of $f_t(t, W(t))$ the Riemann sum converges:

$$\sum_{j=0}^{n-1} f_t(t_j, W(t_j))\Delta t_j \xrightarrow{\|\Pi\|\to 0} \int_0^T f_t(t, W(t))dt$$

$$\sum_{j=0}^{n-1} f_x(t_j, W(t_j))\Delta W_j \xrightarrow{\|\Pi\|\to 0} \int_0^T f_x(t, W(t))dW(t).$$

We can rewrite the third sum as:

$$\sum_{j=0}^{n-1} f_{xx}(t_j, W(t_j))(\Delta W_j)^2 = \sum_{j=0}^{n-1} f_{xx}(t_j, W(t_j))\Delta t_j + \sum_{j=0}^{n-1} f_{xx}(t_j, W(t_j))((\Delta W_j)^2 - \Delta t_j).$$

The first part is a Riemann sum, converging to $\int_0^T f_{xx}(t, W(t))dt$. For the second part, since $(\Delta W_j)^2 - \Delta t_j$ has mean 0 and independent increments, it converges to 0. Hence,

$$\sum_{j=0}^{n-1} f_{xx}(t_j, W(t_j))(\Delta W_j)^2 \xrightarrow{\|\Pi\|\to 0} \int_0^T f_{xx}(t, W(t))dt.$$

Taking limits in (2) yields the claimed result. □

Example 5.4. *Compute $\int_0^T W(t)dW(t)$ using Itô's formula. Let $f(x) = x^2$. Then $f'(x) = 2x$ and $f''(x) = 2$. Applying Itô's lemma with this f gives*

$$W(t)^2 - W(0)^2 = \int_0^t 2W(u)dW(u) + \frac{1}{2}\int_0^t 2\,du.$$

Since $W(0) = 0$ this becomes $W(t)^2 = 2\int_0^t W(u)dW(u) + t$. Rearranging,

$$\int_0^T W(u)dW(u) = \frac{1}{2}(W(T)^2 - T).$$

6 Itô Processes

Definition 6.1 (Itô process)**.** *Let $W(t)$ be a Brownian motion with its natural filtration $(\mathcal{F}_t)$. A stochastic process $X(t)$ is called an Itô process if it can be written as*

$$X(t) = X(0) + \int_0^t \Delta(u)dW(u) + \int_0^t \Theta(u)du,$$

where $X(0)$ is deterministic, $\Delta(u)$ and $\Theta(u)$ are adapted processes, and $\mathbb{E}[\int_0^t \Delta^2(u)du] < \infty$ for all t. Here $\Delta(u)$ is called the volatility and $\Theta(u)$ the drift.

Lemma 6.2. *Let $X(t)$ be an Itô process of the form above. Then its quadratic variation satisfies*

$$[X, X](t) = \int_0^t \Delta^2(u)du.$$

Remark 6.3. *The finite-variation drift term $\int_0^t \Theta(u)du$ contributes no quadratic variation. Only the stochastic integral with respect to W contributes.*

6.1 Itô Formula

Theorem 6.4. *Let $X(t)$ be an Itô process and let $f(t,x) \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R})$. Then*

$$df(t,X(t)) = f_t(t,X(t))dt + f_x(t,X(t))dX(t) + \frac{1}{2}f_{xx}(t,X(t))d[X,X](t).$$

Substituting $dX(t) = \Delta(t)dW(t) + \Theta(t)dt$ and $d[X,X](t) = \Delta^2(t)dt$, we obtain,

$$df(t,X(t)) = f_t(t,X(t))dt + f_x(t,X(t))\Delta(t)dW(t) + f_x(t,X(t))\Theta(t)dt + \frac{1}{2}f_{xx}(t,X(t))\Delta^2(t)dt.$$

6.2 Integral Form of Itô's Formula

Integrating from 0 to t, we obtain

$$f(t,X(t)) - f(0,X(0)) = \int_0^t f_t(u,X(u))du + \int_0^t f_x(u,X(u))dX(u) + \frac{1}{2}\int_0^t f_{xx}(u,X(u))d[X,X](u).$$

Example 6.5 (Generalized Geometric Brownian Motion)**.** *Let $(W(t))_{t\geq 0}$ be a Brownian motion with filtration $(\mathcal{F}_t)$. Let $\alpha(t)$ and $\sigma(t)$ be adapted processes satisfying the usual integrability conditions. Define the Itô process*

$$X(t) = \int_0^t \sigma(s)dW(s) + \int_0^t \left(\alpha(s) - \frac{1}{2}\sigma^2(s)\right) ds.$$

Equivalently, in differential form, $dX(t) = \sigma(t)dW(t) + (\alpha(t) - \frac{1}{2}\sigma^2(t))dt$. Define the asset price process by $S(t) = S(0)e^{X(t)}$, where $S(0) > 0$ is deterministic. Let $f(x) = S(0)e^x$, so $f'(x) = S(0)e^x$ and $f''(x) = S(0)e^x$. Then $S(t) = f(X(t))$, and we may apply Itô's formula:

$$dS(t) = df(X(t)) = f'(X(t))dX(t) + \frac{1}{2}f''(X(t))d[X,X](t).$$

Since $X(t)$ is an Itô process with volatility $\sigma(t)$, $d[X,X](t) = \sigma^2(t)dt$. Substituting,

$$dS(t) = S(0)e^{X(t)}dX(t) + \frac{1}{2}S(0)e^{X(t)}\sigma^2(t)dt = S(t)dX(t) + \frac{1}{2}S(t)\sigma^2(t)dt.$$

Now substitute the expression for $dX(t)$:

$$dS(t) = S(t)\left[\sigma(t)dW(t) + \left(\alpha(t) - \frac{1}{2}\sigma^2(t)\right)dt\right] + \frac{1}{2}S(t)\sigma^2(t)dt = \alpha(t)S(t)dt + \sigma(t)S(t)dW(t).$$

6.2.1 Interpretation

The asset price $S(t)$ satisfies the stochastic differential equation

$$dS(t) = \alpha(t)S(t)dt + \sigma(t)S(t)dW(t)$$

where $\alpha(t)$ is the instantaneous mean rate of return and $\sigma(t)$ is the instantaneous volatility. Both coefficients may be time-varying and random.

Theorem 6.6 (Itô integral of a deterministic integrand)**.** *Let $(W_t)_{t\geq 0}$ be a Brownian motion and let $\Delta : [0,\infty) \to \mathbb{R}$ be a deterministic function such that $\int_0^t \Delta^2(s)ds < \infty$ for every t. Then for each $t \geq 0$,*

$$I(t) \sim \mathcal{N}\left(0, \int_0^t \Delta^2(s)ds\right).$$

Remark 6.7. *Please refer to Shreve for a proof of this. To show mean is 0, we use that the Itô integral is a martingale and the variance follows from the quadratic variation of an Itô integral.*

6.3 Example: Cox-Ingersoll-Ross (CIR) interest rate model

Example 6.8. *Let $(W_t)_{t\geq 0}$ be a Brownian motion. The Cox-Ingersoll-Ross (CIR) model for the (short) interest rate $(R_t)_{t\geq 0}$ is*

$$dR(t) = (\alpha - \beta R(t))dt + \sigma\sqrt{R(t)}dW(t), \quad \alpha, \beta, \sigma > 0. \quad (3)$$

Consider the function $f(t,x) = e^{\beta t}x$ and apply Itô's formula to $f(t,R(t)) = e^{\beta t}R(t)$. We have $f_t(t,x) = \beta e^{\beta t}x$, $f_x(t,x) = e^{\beta t}$, $f_{xx}(t,x) = 0$, hence

$$\begin{aligned} d(e^{\beta t}R(t)) &= f_t(t,R(t))dt + f_x(t,R(t))dR(t) + \frac{1}{2}f_{xx}(t,R(t))dR(t)dR(t) \\ &= \beta e^{\beta t}R(t)dt + e^{\beta t}[(\alpha - \beta R(t))dt + \sigma\sqrt{R(t)}dW(t)] \\ &= \alpha e^{\beta t}dt + \sigma e^{\beta t}\sqrt{R(t)}dW(t). \end{aligned}$$

Integrating from 0 to t gives

$$e^{\beta t}R(t) = R(0) + \alpha\int_0^t e^{\beta u}du + \sigma\int_0^t e^{\beta u}\sqrt{R(u)}dW(u) = R(0) + \frac{\alpha}{\beta}(e^{\beta t} - 1) + \sigma\int_0^t e^{\beta u}\sqrt{R(u)}dW(u). \quad (4)$$

Taking expectations,

$$e^{\beta t}\mathbb{E}[R(t)] = R(0) + \frac{\alpha}{\beta}(e^{\beta t} - 1),$$

and hence,

$$\mathbb{E}[R(t)] = e^{-\beta t}R(0) + \frac{\alpha}{\beta}(1 - e^{-\beta t}). \quad (5)$$